

A13 — Robotic Design and Ethical Analysis Project

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1. Application Domain

Domestic/industrial assistance — commercial HVAC & pharmaceutical cleanroom service.

This domain is chosen because it's real, lived experience — commercial HVAC/refrigeration work (EPA 608 Universal, TDLR #148984) and cGMP biotech cleanroom environments, not a hypothetical.

2. Robot Design

Name: *SentryTech* — a pre-dispatch verification and cleanroom environmental-monitoring robot.

Description: A wheeled, waist-height service robot that does two jobs depending on deployment context:

1. **Commercial HVAC pre-dispatch verification** — before a technician is dispatched to a job site, SentryTech (or a stationary version mounted at the unit) scans the rooftop unit's current state — fault codes, visible parts/tools staged nearby, access panel condition — and cross-references it against the work order. This directly targets a real failure mode: technicians arriving on-site with the wrong tools/parts because the work order didn't match actual field conditions (the kind of mismatch that causes wasted trips).
2. **cGMP cleanroom environmental monitoring** — in a pharma/biotech cleanroom, a stationary/rail-mounted version continuously monitors particle counts, temperature, humidity, and differential pressure, flagging deviations before they become a batch-compliance incident.

Physical appearance: Compact, sealed housing (IP-rated for cleanroom/HVAC-adjacent environments), a fixed camera arm, and a small diagnostic display. No manipulator arm — this robot observes and reports, it does not physically intervene.

3. Functionality and AI Integration

- **Computer vision** compares the live camera feed against the work order/spec sheet — e.g., confirming a tool or part described in the order is actually staged, or that a cleanroom's visible conditions (spill, open panel) don't already violate protocol.

- **Fault-code parsing** — reads a unit's digital fault/status codes and flags mismatches between reported state and dispatch paperwork.
- **Anomaly detection on sensor time-series** (cleanroom mode) — particle count, temp, humidity, pressure readings run through a model trained to flag deviations from the room's established baseline, not just fixed thresholds — because "normal" varies by room and season.
- **Autonomy boundary:** the robot never makes a go/no-go call on its own. It surfaces a flag; a human technician or QA lead makes the actual decision. This is a deliberate design constraint, not a limitation to apologize for.

Sensors/inputs: camera(s), particle counter, temperature/humidity/ differential-pressure sensors, and a read interface for the HVAC unit's existing fault-code system (no new hardware installed on the unit itself).

4. Ethical Analysis

Privacy: Camera-based monitoring in a workspace raises the standard workplace-surveillance question — is this observing the technician, or the equipment? Mitigation: field of view and recording scope are contractually and technically restricted to equipment/workspace state, not worker behavior; footage retention is short and audit-logged.

Safety: A false "all clear" from the AI (e.g., missing a real fault code) could send a technician into a hazardous cleanroom or job site under wrong assumptions. Mitigation: the system is explicitly advisory — it flags for human review, and standard PPE/protocol checks remain mandatory regardless of what the robot reports.

Job displacement: This tool doesn't replace a technician's diagnostic judgment — it replaces the wasted trip, not the trip itself. Worth naming honestly: pre-dispatch verification could reduce dispatch *volume* even as it protects the quality of dispatches that do happen; that tradeoff deserves to be stated, not glossed over.

Societal impact: In cGMP contexts specifically, false negatives (missed deviations) have downstream consequences for drug safety, not just operational efficiency — the ethical stakes scale with the industry the robot is deployed into.

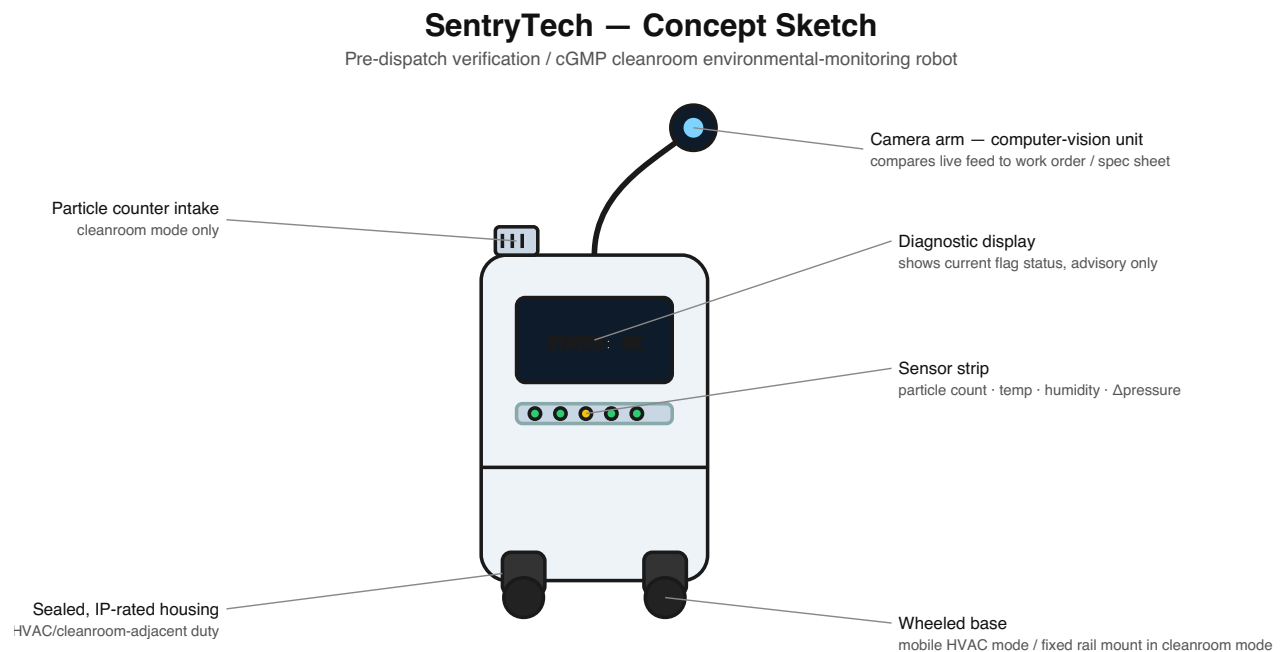
Proposed guidelines: - Human sign-off required before any dispatch decision the AI flags. - Camera footage scoped to equipment/workspace, not worker monitoring. - Regular audit of false-negative rate in cleanroom mode, given the compliance stakes.

5. Research Component

This design and ethical analysis draws on the following foundational and current sources:

- Murphy, R. R. (2000). *Introduction to AI Robotics*. MIT Press. — foundational principles behind sensor-driven autonomy and the perception-action loop used in SentryTech's design.
- Jobin, A., Ienca, M., & Vayena, E. (2019). The global landscape of AI ethics guidelines. *Nature Machine Intelligence*, 1, 389–399. — informs the privacy/safety/accountability framework applied in Section 4.
- UNESCO. (2021). *Recommendation on the Ethics of Artificial Intelligence*. — grounds the human-sign-off and proportionality principles in the proposed guidelines.
- Khan, A. A., Badshah, S., Liang, P., et al. (2022). Ethics of AI: A systematic literature review of principles and challenges. In *Proceedings of the International Conference on Evaluation and Assessment in Software Engineering (EASE)*. arXiv:2109.07906. — supports the job-displacement and societal-impact discussion.

6. Sketch



Two Deployment Modes (same hardware, different config)

1. HVAC pre-dispatch verification — mobile unit scans RTU fault codes + staged parts/tools against the work order before a technician is dispatched.
2. cGMP cleanroom monitoring — rail-mounted, continuously tracks particle count, temp, humidity, and differential pressure against the room's established baseline.

Autonomy boundary: the robot flags for human review — it never issues a go/no-go decision on its own.

Digitally created concept sketch — labeled diagram showing the camera arm (computer vision), diagnostic display, sensor strip (particle count, temperature, humidity, differential pressure), sealed IP-rated housing, and wheeled base, with both deployment modes (HVAC pre-dispatch / cGMP cleanroom) noted below.

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